

Nisa Nabilah Azaddin

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Data Science and AI undergraduate with hands-on research and enterprise experience delivering machine learning solutions. First-author publication at IEEE ICT4M 2025, with applied work spanning graph neural networks (GraphSAGE), NLP pipelines, and cloud-native ML systems. Proficient across the full data science stack (Python, PyTorch, SQL, PySpark, Power BI, Google Cloud Platform). Passionate about turning complex data into clear, actionable decisions.

EXPERIENCE

AI Intern STDCx, Teraju AI Internship Programme | 2026 - Present

- Successfully developed ClearDesk, a browser-based workplace-protection system featuring SHA-256-sealed audit trails, AI-powered scope-creep detection, and append-only incident logging, to be deployed as a functioning enterprise tool.
- Established a self-initiated AML fraud detection pipeline using GraphSAGE (PyTorch Geometric) on the Elliptic Bitcoin Dataset, achieving Test F1 0.7845 and AUC-PR 0.9857, with a Streamlit + PyVis dashboard and GNNExplainer-based model interpretability.
- Assembled a Telegram-integrated attendance automation system synced with Google Sheets, reducing manual tracking time by ~3 hours/week across a 20-person cohort.
- Liaised across three concurrent AI initiatives (attendance automation, document intelligence, model monitoring), translating business requirements into ML task specifications for delivery teams.

Research Assistant, MOSTI-Funded Enterprise AI Project: Forthify Tech (OMIS, IIUM) | 2025

- Conducted end-to-end NLP pipeline development (tokenisation, NER, text classification) using Python and HuggingFace Transformers, consistently achieving 88% classification accuracy on enterprise datasets.
- Reduced model error rate by ~12% through iterative feature engineering, hyperparameter optimisation, and cross-validation, validated via formal precision/recall/F1 benchmarking across model iterations.
- Facilitated cross-functional stakeholder alignment, translating requirements into AI deployment specifications and authoring system behaviour documentation that anchored engineering handoff.

ADDITIONAL EXPERIENCE

Personal Assistant / Scheduling Coordinator | 2025

- Coordinated and managed end-to-end scheduling across 5 concurrent high-stakes commitments for a multi-venture founder.

Freelance Brand Promoter & Event Crew | 2023–2025

- Represented four national brands (GPR, Dutch Lady, Yobick, MATTA) across live activations and campaigns.

Senior Sales Assistant, AS Watson | 2021

- Consistently achieved sales KPIs while responsible for customer service, merchandising, and stocktaking in a high-volume retail environment.

EDUCATION

Bachelor In Computer Science (Honours) (Data Science and Artificial Intelligence)

International Islamic University Malaysia (IIUM) | CGPA: 3.40 / 4.00 | Expected 2026

- Dean's List: 3 semesters (2023/2024 Sem 1, 2024/2025 Sem 1, 2025/2026 Sem 1)

TECHNICAL SKILLS

Languages: Python, SQL, R, Java, HTML, CSS, C

ML & Deep Learning: PyTorch, PyTorch Geometric, GraphSAGE, Scikit-learn, HuggingFace Transformers, NLP, Predictive Modelling, Feature Engineering

Data & Cloud: BigQuery, PySpark, Google Cloud (Vertex AI), Git/GitHub, Streamlit
Analytics & Visualization: Power BI, Looker Studio, EDA, Statistical Analysis

CERTIFICATIONS

- Google Cloud Skills Badges: BigQuery, Looker Studio, Vertex AI, Cloud Security
- CompTIA Data+ ce (DAO-001)
- CCNA: Introduction to Networks (Cisco)

PUBLICATIONS

IEEE ICT4M 2025 | First Author

Quantitative Imaging Biomarkers for Prognosis Prediction and Treatment Stratification in Lung Cancer Through a Multi-Stage Deep Learning Approach

PROJECTS

Code-Switch NLP Classifier

Malaysian English–Malay Code-Switching Detection | Python, Scikit-learn, HuggingFace Transformers, mBERT

- Built a multilingual text classifier to detect English–Malay code-switching in Malaysian social media discourse, engineering a synthetic dataset that captured authentic bilingual linguistic patterns.
- Benchmarked three modeling approaches, TF-IDF + Logistic Regression, TF-IDF + Linear SVM, and fine-tuned multilingual BERT (mBERT) with mBERT consistently outperforming classical baselines on mixed-language sentence classification.
- Delivered a regionally-specific NLP solution addressing a linguistic challenge underrepresented in mainstream NLP datasets, demonstrating applied cross-lingual model fine-tuning.

AML Detection via Graph Neural Networks

Python, PyTorch Geometric, GraphSAGE, GNNExplainer, Streamlit, PyVis

- Trained a GraphSAGE model on the Elliptic Bitcoin Dataset (203K nodes, 234K edges), achieving a test F1 of 0.7845 and AUC-PR of 0.9857, surpassing the Weber et al. benchmark (0.88).
- Conducted temporal analysis across all 49 time steps to assess model stability under distribution shift, and applied GNNExplainer for subgraph-level interpretability.
- Built an interactive Streamlit and PyVis dashboard for transaction graph exploration.

LEADERSHIP & VOLUNTEER EXPERIENCE

Secretary I / Core Organizing Committee, Motion-U Startup Incubation Club | 2022–2025

- Was elected to Secretary I and successfully revived a defunct club from zero, establishing the MotionU Award, a competitive FYP-style showcase now part of the club's flagship programming.
- Was appointed Chair for 3 consecutive years, coordinating 5+ major initiatives and engaging 70+ members and external industry partners.

Event Volunteer/Support

- MyHackathon (Cradle Fund & MOSTI), "Anonymity 101" Cybersecurity Talk (ICE)

PROFESSIONAL DEVELOPMENT

- **CelcomDigi SiswaPreneur Bootcamp (2023)** | Platform Strategy, Zero-Cost Marketing, Online Presence Building, AI-Driven Content Creation
- **Celcom Siswapreneur @ Google Bootcamp (2022)** | AI in Daily Life, Google Collaboration Tools, Personal Branding, Career Readiness